Oscar Turner

Curriculum Vitae

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EDUCATION

Haverford College — Bachelor of Science in Physics & Astronomy and Bachelor of Arts in Philosophy
Expected: May 2029
GPA: 3.93/4.00
Work 20 hours/week across 4 campus jobs on top of a full course load and research.
Relevant coursework: Cosmology, Waves and Optics, Extragalactic Data Science, Real Analysis, Differential Equations, Astrophysics, Electricity and Magnetism.

Hunter College High School
June 2025
GPA: 97/100
Advanced diploma with departmental honors in physics, math, and English, and the Lilian H. Sigler award given to one student for achievement in the humanities. Hunter College High School represents the top one quarter of one percent of students in New York City based on test scores.

RESEARCH EXPERIENCE

Galaxy Quenching in the MaNGA Survey

May 2026-Present
Haverford College
Advised by Prof. Karen L. Masters and Dr. Kelly Sanderson
I work on low-ionization emission regions, tracers of the end of star formation in galaxies, and use resolved maps of these regions from the MaNGA data to analyze patterns and dynamics of galaxy quenching in a sample of 10,000+ galaxies. Developed a Python code to classify the full MaNGA sample based on ionization morphology and performed data analysis on this classified sample. We find evidence for complex processes of quenching dependent on stellar mass and environment, with different mechanisms active at different galactic radii. Paper in prep.

Analysis of Supernova Candidate AT2024nzx

June-August 2024
Yale University; Yale Summer Program in Astrophysics
Advised by Dr. Michael Faison and Mr. Michael Warrener
One of 32 high school juniors selected out of 1000+ international applicants for a 6-week research intensive at Yale. Performed nightly observation of an unusual supernova candidate located near the galactic plane, which typically obscures supernovae. Worked in a team of 4 students to analyze the transient, performing data analysis and eventually reclassifying it as a cataclysmic variable. Learned Python, AstroImageJ, Siril, and telescope operation techniques equivalent to a semester course in observational astronomy and data science.

TEACHING EXPERIENCE

Teaching Assistant

Physics and Astronomy Department, Haverford College Courses: Electricity and Magnetism, Astrophysics
Spring 2026 - Present
Lead 9 hours/week of physics clinic in introductory courses in electricity and magnetism (Spring 2026) and mechanics (Fall 2027). Assist 20+ students/week with homework and exam review. Grade weekly homework assignments for a class of 50+.

Writing Center Tutor

Writing Center, Haverford College
Fall 2027 - Present
Weekly one-on-one mentorship of incoming freshmen writers.

AWARDS

Brian Kovaric Scholarship
Awarded by Haverford College, 2026
Summer lab placement and funding, awarded to two members of the freshman class who demonstrate "exceptional potential for science research."

Silver Medal for a Writing Portfolio
Awarded by the Scholastic Art and Writing Awards, 2025
$2000 scholarship awarded to 12 creative writing portfolios out of 7000+ submissions by high school seniors from across the country. My essay "Going Downhill" was 1 of 4 writing submissions out of 350,000+ total to the awards selected to be read aloud at Carnegie Hall for an audience of 1000+.

PUBLICATIONS

Journal Articles

A paper in prep. on galaxy quenching.

Conference Proceedings

Quantifying Quenching in MaNGA Galaxies
September 2026
Proceedings of the Keck Northeast Astronomy Consortium.

CONFERENCE ACTIVITY

Talks Presented

Resolved Star Formation Properties in the MaNGA Sample
September 2026
Invited talk to the Galaxy Zoo team.

Quenching in MaNGA Galaxies
September 2026
Annual Meeting of the Keck Northeast Astronomy Consortium - selected talk.

Quantifying Quenching: cLIERs, eLIERs, and outLIERs in MaNGA Galaxies
July 2026
Philadelphia Kpc+ Astrophysics and Theory Conference - selected talk.

SERVICE & OTHER ACTIVITY

Honor Councilor

Fall 2025 - Present
Haverford's Honor Code is the oldest student-run code in the country, and students are given complete agency over enforcement of academic integrity at the school. I was elected to represent the class of 2029 on Honor Council, serving on juries for academic integrity cases, working to improve transparency, addressing AI use on campus, and advancing diversity initiatives. In Fall 2025, I was part of a committee which rewrote the complete honor code, including the social honor code, which had been subject to legal challenges from the federal government. I presented the revised code to the student body, helping secure its adoption.

Other activities and interests

On campus, I work as a manager of Haverford's communal kitchen, cleaning it daily and handling a $1000 annual budget for ingredients and supplies. As a creative writer, I'm involved with Haverford's literary magazine, and I've previously worked with the BreakBread Literacy Project to edit a national literary magazine for young, underprivileged writers. My other interests include philosophy of physics, philosophical psychology, classical and jazz piano (I've played for 10 years), camping, literature, and meditation.